STA 013 — Problem Set (Day 2)

Topic: Dotplots, Stem-and-Leaf, Histograms & Intro to Probability | Lecture: June 23 (Professor Max)

Instructions: *Choose the best answer for each multiple-choice question. Several questions require you to construct or read values from a histogram — show your work where calculations are needed.*

1. A professor collects the following raw values during a demo: 3, 5, 5, 8. No context about where these values came from is given. Which statement is correct?

A) This is a sample, because it has more than one data point.

B) This is a population, because all values were recorded.

C) This is neither a sample nor a population — it is simply data, since no population of interest has been defined.

D) This is both a sample and a population simultaneously.

2. Construct a stem-and-leaf plot for the following data: 42, 58, 45, 61, 58, 47, 60. After sorting and splitting into stems (tens digit) and leaves (ones digit), which stem has the MOST leaves?

A) Stem 4

B) Stem 5

C) Stem 6

D) All stems are tied

3. In a stem-and-leaf plot built from real data, one stem value has zero leaves (no data fell in that range), but the stem is still listed on the plot with nothing next to it. Why?

A) It's a typo and should be removed

B) Listing it preserves the true shape and spacing of the distribution; omitting it would compress the gap and misrepresent the data

C) Every stem-and-leaf plot is required to have exactly the same number of stems regardless of the data

D) Empty stems indicate the data must be qualitative, not quantitative

4. A histogram has a single, clear peak, and the data trails off gradually to the right with most values clustered on the left. How should this distribution be described?

A) Unimodal and left-skewed

B) Unimodal and right-skewed

C) Bimodal and symmetric

D) Multimodal and right-skewed

5. A data point sits noticeably apart from the rest of a data set, but no one has yet investigated why. What is the most statistically correct way to refer to this point right

now?

A) A confirmed outlier

B) A potential outlier

C) A mutually exclusive event

D) A bimodal value

6. A quality control technician records 16 gear diameters (in cm) and inadvertently makes a typing error on one entry:

1.991, 1.891, 1.991, 1.988, 1.993, 1.989, 1.990, 1.988, 1.988, 1.993, 1.991, 1.989, 1.989, 1.993, 1.990, 1.994

Which value is the potential outlier, and what is the most likely explanation?

A) 1.994 — the machine was running too fast

B) 1.891 — likely a typing/data-entry mistake, since it falls far outside the tight cluster of the other values

C) 1.988 — it appears most frequently, so it must be the error

D) There is no outlier; all values are part of the natural spread

7. A data set of 50 ages ranges from a minimum of 25 to a maximum of 73. The professor asks you to construct a relative frequency histogram using 6 bins. What is the bin width, and what are the bin boundaries (starting at the minimum)?

A) Bin width = 6; boundaries: 25, 31, 37, 43, 49, 55, 61

B) Bin width = 8; boundaries: 25, 33, 41, 49, 57, 65, 73

C) Bin width = 9.6; boundaries: 25, 34.6, 44.2, 53.8, 63.4, 73

D) Bin width = 48; boundaries: 25, 73

→ Hint: Bin Width = (Max – Min) / #Bins, then add the bin width repeatedly starting from the minimum value.

8. Using left-inclusive bins of [33, 41) and [41, 49), where would a data value of exactly 41 be placed?

A) In the [33, 41) bin, since 41 is the upper boundary

B) In the [41, 49) bin, since left-inclusion means the left endpoint of an interval is included in that interval

C) In both bins simultaneously

D) It would be excluded from the histogram entirely

9. A relative frequency histogram has 4 bins, each of width 5, with the following relative frequencies: [0,5): 0.10, [5,10): 0.35, [10,15): 0.40, [15,20): 0.15. What is the probability that a randomly selected measurement is AT LEAST 10?

A) 0.40

B) 0.45

C) 0.55

D) 0.75

→ Hint: “At least 10” includes the [10,15) bin AND the [15,20) bin.

10. Professor Max asks: “If we roll a die, what is the probability of rolling a 4?” Without any other information given, what is the correct response?

A) 1/6, since all dice are fair by default

B) We cannot determine the exact probability, because we don't know if the die is fair (equally likely outcomes are not guaranteed)

C) 0, since rolling a 4 is impossible

D) 1/4, since there are 4 faces being considered

11. A student has data from a SAMPLE of 200 customers (not the full customer population) and wants to estimate how often customers choose a particular product. Which concept should the student technically use to describe this estimate?

A) Probability, since any data set can use probability

B) Relative frequency, since this is a sample rather than the full population

C) Mutual exclusivity, since only one product can be chosen at a time

D) Sample space, since 200 is the total number of outcomes

12. A biased coin has P(Heads) = 0.4 and P(Tails) = 0.6. The coin is tossed twice. What is the probability of observing AT LEAST one head?

A) 3/4, same as the fair coin case

B) 0.16, since P(H) × P(H) = 0.4 × 0.4

C) 0.64, found by computing 1 – P(no heads) = 1 – (0.6 × 0.6)

D) 0.40, since that's the probability of heads on a single toss

→ Hint: With an unfair coin, you cannot simply count outcomes out of 4 equally likely possibilities — you must multiply actual probabilities. P(at least one head) = 1 – P(TT).

Answer Key & Explanations

1. C — Data only becomes a “sample” once a population of interest has been defined; on its own, it's just a set of measurements. This directly echoes the professor's repeated point in lecture.

2. A — Sorted: 42, 45, 47, 58, 58, 60, 61. Stem 4 → leaves 2, 5, 7 (3 leaves). Stem 5 → leaves 8, 8 (2 leaves). Stem 6 → leaves 0, 1 (2 leaves). Stem 4 has the most leaves.

3. B — Listing empty stems preserves the true shape/spacing of the distribution; skipping them would visually compress gaps and misrepresent how the data is distributed.

4. B — A single peak = unimodal. Data clustered on the left with a tail trailing to the right = right-skewed (right tail).

5. B — Until the cause of separation has been investigated, statistical convention calls it a “potential outlier,” not a confirmed one.

6. B — 1.891 sits far outside the tight cluster of values between 1.988–1.994, consistent with a data-entry typo (most likely meant to be 1.991).

7. B — Bin width = (73–25)/6 = 48/6 = 8. Boundaries starting at 25, adding 8 repeatedly: 25, 33, 41, 49, 57, 65, 73.

8. B — Left-inclusion means the left endpoint of an interval belongs to that interval. So 41 belongs to [41, 49), not [33, 41).

9. C — “At least 10” means 10 and above: 0.40 + 0.15 = 0.55.

10. B — Without confirmation the die is fair, we cannot assume equally likely outcomes, so the exact probability is unknown — this is the exact wording trap the professor warned about.

11. B — Since this is a sample (not the full population of customers), the correct concept is relative frequency (f/n), not probability.

12. C — P(at least one head) = 1 – P(no heads) = 1 – P(TT) = 1 – (0.6 × 0.6) = 1 – 0.36 = 0.64. With an unfair coin, outcomes are NOT equally likely, so you must multiply actual probabilities rather than counting outcomes out of 4.